Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/03/2006
PI Name:	Long, Marshall Ph.D.		
Project Title:	Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: end date should be 9/30/2006; grants reinstatement resulted in new grant; see project with same title appended with new grant number--NNX07AB33A--per D. Stocker/GRC (10/07) NOTE: grant reinstated 8-06 through 9/30/07, per J. Cohen (HQ). Task not in Task Book prior to 8/2006		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smooke, Mitchell David (Yale University)		
Grant/Contract No.:	NNC04AA03A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To better understand the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum - namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames which have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform from which to better understand the factors that affect diffusion flame lift-off distance and extinction and it should also provide an excellent test-bed to determine the roles of inception, surface growth and oxidation processes in soot formation models The goals of the research will be the generation of modified kinetic mechanisms for hydrocarbon fuels that are able to model effectively diffusion flame structure under a larger parameter range than existing mechanisms and the development of submodels for soot formation that are capable of preditcting both high and low soot loading levels in hydrocarbon flames of various fuels (methane, ethylene and propane).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation and radiative transfer take place.
Task Progress:	During the initial stages of our project, we continued the analysis of the data obtained in our previous microgravity work, which concentrated on the lift-off behavior of our coflow diffusion flames in both normal and microgravity environments. More recently, our experimental work has centered on the development of measurement techniques that use simple experimental configurations to study sooting tendencies of flames at normal gravity. These techniques can later be applied to measurements made in a microgravity environment. The focus of the computational work has been on improving the modeling of soot formation. Some of the details of these advances are given in the following paragraphs. Experimentally, we have improved and extended our work using a digital camera (the Nikon D70) as an optical pyrometer. Use of the built-in color filter array (CFA) of a digital camera allows for two-dimensional imaging of flame emission spectra at the wavelengths of the red, green and blue color filters, without the need to spatially match images taken at the different wavelengths. As a result, it is possible to determine soot temperature and concentrations using a single picture. Previously, the filter profiles of the CFA were characterized to provide a calibration for our instrument. Experimentally determined temperatures were found to be approximately 100 K higher than those calculated numerically. Though the data agreed with expected trends, it was difficult to conclude on the overall accuracy of the temperatures measured, as comparison to a source of known temperature was not available. Currently, an independent blackbody calibration is under development. Images have been taken of an oven at varying temperatures. In order to check the consistency of the two independent calibration techniques, temperature values for the data from the oven were determined using the CFA calibration. The calculated temperatures were found to be 94 K higher, on average, than the temperature that was set on the oven. A sensitivity analysis of the CFA calibration, which depends on a number of different measurements, shows the determination of the measured filter profiles to be critical to the final temperatures calculated, e.g., a shift of the red-channel's central wavelength of 5 nm corresponds to change in calculated temperature of 50 K. Therefore, a blackbody calibration can be considered more robust than a calibration of the camera's CFA. Preliminary results from the blackbody calibration for ethylene flames of different fuel concentrations (32%, 40%, 60%, and 80% ethylene, diluted with nitrogen) are found to have the same qualitative behavior as with the CFA calibration, with temperatures from the blackbody calibration being almost 100 K higher overall. Once the local temperatures have been determined, the blackbody calibration makes it possible to estimate the soot volume fraction of the test flames using the pyrometer data. Information of the radial signal intensity at a given color channel, coupled with the experimentally determined local temperature are used to calculate the local soot concentration. The resulting soot concentrations from the pyrometer data can then be compared to soot volume fraction measurements made previously using laser-induced incandescence (LII). Peak soot concentrations, as well as variations in concentration across each flame, show excellent agreement between both measurement techniques. Progress in our LII measurements has been made through implementation of an on-line extinction calibration, which provided an improved optical calibration to the system. In addition, the use of a fast camera system removed a measurement bias towards larger particles. The use of this fast camera, which can be gated down to 5 ns, makes two-dimensional particle-sizing measurements possible by determining the cooling rate of particles that have been heated by the laser. This particle-sizing technique and further improvements to the LII system are currently under development. A recent LII workshop provided the opportunity for researchers around the world to discuss this measurement technique in order to provide consensus as to the best-practices, and to try and eliminate discrepancies in implementation that occur between different research groups. Conclusions and inputs acquired from this meeting will be added to our LII system. Computationally, soot kinetics are modeled as particles consisting of isolated spheroids, which undergo surface growth and oxidation and coalesce with one another on impact. The particle mass range of interest is divided into sections and an equation is written for each section including coalescence, surface growth, and oxidation. For the smallest section, an inception source term is included. The transport conservation equation for each section includes thermophoresis, an effective bin diffusion rate, and source terms for gas-phase scrubbing. The gas and soot equations are additionally coupled through non-adiabatic radiative loss in the optically-thin approximation. In previous studies, as the fuel dilution increases, we have predicted the migration of the high soot regions from the centerline of the flame to the wings. To understand this behavior more completely, we have begun a detailed investigation of the relative rates of inception, surface growth, and oxidation, along with a particle residence time analysis, to explain the shift of soot away from the

centerline as the fuel fraction increases. While inception tends to peak on the centerline, the maximum in surface growth migrates from the centerline to the wings of the flame as the fuel fraction increases. Concurrently, the relative importance of surface growth and inception reverses. This change in the relative importance of these two subprocesses is due to the significant increase in residence time available for soot growth in the flame wings. By altering the inlet velocities of the higher fuel fraction flame, we are able to modify the available residence time along the wings so as to generate soot profiles similar to those of a lower fuel fraction flame. The level of soot on the centerline increases with respect to that on the wings as the peak centerline to wing temperature also increases.

In modeling the growth of soot particles, it has been conventional to assume that the particles consist of isolated spheroids, which undergo surface growth and oxidation and coalesce with one another on impact. The experimental evidence, however, is that while the coalescing spheroid picture is accurate for very small, young, particles, older soot particles tend to consist of chainlike aggregates of primary spheroids which fuse together on impact without coalescing. The aggregation process is important in soot formation because it affects the particle surface area available for growth and oxidation. Further, the primary particles within aggregates do not grow indefinitely, but tend to reach a limited maximum size not strongly dependent on either the flame or the fuel. To have a more realistic description of particle structure, the sectional aerosol dynamics representation that has been used to model the growth of spheroidal aerosols needs to be extended to model soot aggregates, as well.

To remedy these deficiencies, we are in the process of employing a discrete-sectional aerosol dynamics model, which has the feature of predicting aggregate growth. The algorithm has provision for a “discrete” size range of precursor or nascent particles whose masses are multiples of some inception growth species. The discrete size range is joined to an isolated spheroid size range whose boundaries vary linearly on a logarithmic scale as in the conventional sectional model; these “liquid” particles are assumed to coalesce on collision. Beyond an arbitrarily determined size, particles are assumed to fuse with one another, leading to formation of chainlike aggregates of smaller primary particles. The conventional spheroid sectional analysis tracks the total mass density within set sectional boundaries; the extended sectional analysis tracks not only the mass density, but also the number of primary particles within an aggregate section. The transition from liquid to solid particles occurs at a particle diameter that is an input item, and the mass fractal dimension of the aggregates is also assumed. Aggregate-aggregate collisions leading to larger aggregates are prescribed at rates based on the assumed fractal dimension.

Once the model is completed and fully implemented, we will carry out a set of computations first on normal gravity flames for which we have an extensive set of LII measurements and then on zero-g flames.

Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Smooke MD, Long MB, Connelly BC, Colket MB, Hall RJ. "Soot formation in laminar diffusion flames." Combustion and Flame. 2005 Dec;143(4):613-28. http://dx.doi.org/10.1016/j.combustflame.2005.08.028 , Dec-2005
Articles in Peer-reviewed Journals	Dworkin SB, Connelly BC, Schaffer AM, Long MB, Smooke MD, Puccio MP, McAndrews B, Miller JH. "Computational and experimental study of a forced, time-independent, methane-air coflow diffusion flame." Proceedings of the Combustion Institute. 2007 Jan;31(1):971-8. http://doi.org/10.1016/j.proci.2006.08.109 , Jan-2007
Papers from Meeting Proceedings	Connelly BC, Bennett BA, Dworkin SB, Smooke MD, Long MB, Puccio MA, Herdman JD, Miller JH. "Computational and Experimental Study of Molecular Growth in Forced, Time-Varying Flames." Eastern States Section of the Combustion Institute, University of Central Florida, Orlando, FL, November 2005. Proceedings of the Eastern States Section of the Combustion Institute, University of Central Florida, Orlando, FL, November 2005. , Nov-2005